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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H1 CHEMISTRY

8872/01

Paper 1 Multiple Choice

20 September 2012
50 minutes

Additional Materials: Optical Mark Sheet
 Data Booklet

INSTRUCTIONS TO CANDIDATES

1. Write your name, index number and class on this question paper.
2. There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Optical Mark Sheet.
3. Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
4. Any rough working should be done in this booklet.
5. You may use a calculator.

This question paper consists of **13** printed pages and **1** blank page.

Section A

For each question, there are four possible answers **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1** Which mass of gas would occupy a volume of 3 dm³ at 25 °C and 1 atm.

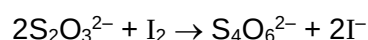
A 3.2 g O₂ gas
B 5.6 g N₂ gas
C 8.0 g SO₂ gas
D 11.0 g CO₂ gas

- 2** Use of the *Data Booklet* is relevant to this question.

2.920 g of a Group II metal, **X**, reacts with an excess of chlorine to form 5.287 g of a metal chloride. What is metal **X**?

A barium
B calcium
C magnesium
D strontium

- 3** When an excess of KI solution was added to 10.0 cm³ of a 0.200 mol dm⁻³ solution of JO₄³⁻, the iodine liberated was titrated against a standard solution of 0.100 mol dm⁻³ of sodium thiosulfate, Na₂S₂O₃, using starch as the indicator.



If the volume of sodium thiosulfate required to discharge the blue–black colouration is 40.00 cm³, what is the final oxidation state of **J**?

A –3 **B** +1 **C** +2 **D** +3

- 4** How many moles of electrons must be removed from each mole of methylbenzene, C₆H₅CH₃, when it is oxidised to benzoic acid, C₆H₅COOH?

A 1 **B** 2 **C** 4 **D** 6

- 5** Use of the *Data Booklet* is relevant to this question.

What could be the proton number of an element that has three unpaired electrons in each of its atoms?

A 5 **B** 13 **C** 15 **D** 21

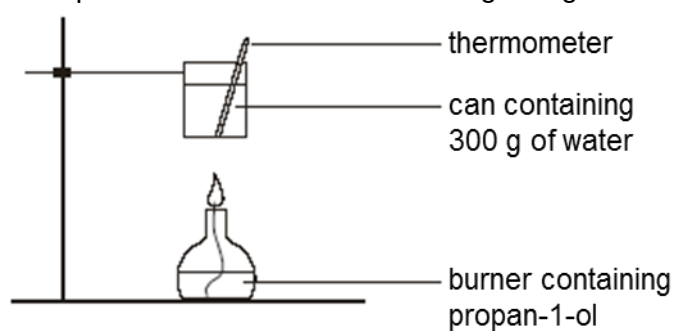
- 6 Use of the *Data Booklet* is relevant to this question.

The elements radon (Rn), francium (Fr) and radium (Ra) have consecutive proton numbers in the Periodic Table.

What is the order of their first ionisation energies?

	least endothermic	→	most endothermic
A	Fr	Ra	Rn
B	Fr	Rn	Ra
C	Ra	Fr	Rn
D	Rn	Ra	Fr

- 7 A student used the set-up below to heat a can containing 300 g of water.



The following data were recorded:

$$\begin{aligned}\text{mass of propan-1-ol burnt} &= m \text{ g} \\ \text{change in temperature of water} &= \Delta T ^\circ\text{C}\end{aligned}$$

Given that:

$$\begin{aligned}\text{relative molecular mass of propan-1-ol} &= 60.0 \\ \text{enthalpy change of combustion of propan-1-ol} &= -2021 \text{ kJ mol}^{-1} \\ \text{specific heat capacity of water} &= c \text{ J g}^{-1} \text{ K}^{-1}\end{aligned}$$

What is the efficiency of this heating process?

A $\frac{m \times 2021 \times 1000}{300 \times c \times \Delta T \times 60.0} \times 100\%$

B $\frac{m \times c \times \Delta T \times 60.0}{300 \times 2021 \times 1000} \times 100\%$

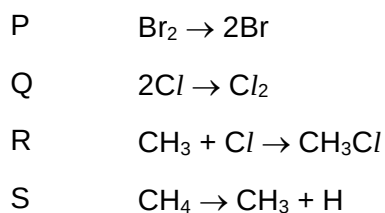
C $\frac{300 \times c \times \Delta T \times 60.0}{m \times 2021} \times 100\%$

D $\frac{300 \times c \times \Delta T \times 60.0}{m \times 2021 \times 1000} \times 100\%$

- 8 Some bond energy values are listed below.

bond	bond energy / kJ mol^{-1}
C-H	410
C-Cl	340
Cl-Cl	244
Br-Br	193

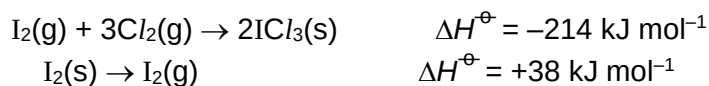
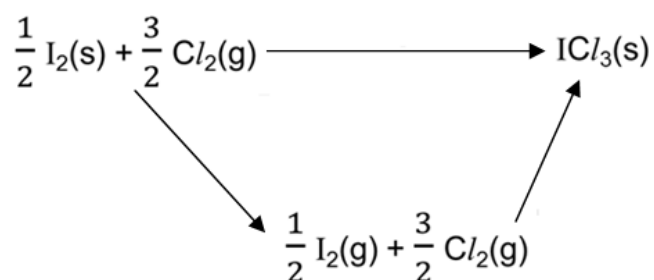
These bond energy values relate to the following four reactions.



What is the order of enthalpy changes of these reactions from most negative to most positive?

- A $\text{P} \rightarrow \text{Q} \rightarrow \text{R} \rightarrow \text{S}$
 B $\text{Q} \rightarrow \text{R} \rightarrow \text{S} \rightarrow \text{P}$
 C $\text{S} \rightarrow \text{P} \rightarrow \text{Q} \rightarrow \text{R}$
 D $\text{R} \rightarrow \text{Q} \rightarrow \text{P} \rightarrow \text{S}$

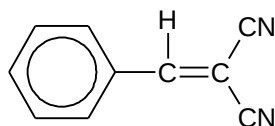
- 9



Using the information above, calculate the standard enthalpy change of formation of iodine trichloride, $\text{ICl}_3(\text{s})$.

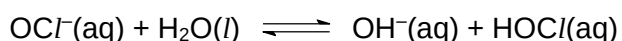
- A $+176 \text{ kJ mol}^{-1}$
 B -88 kJ mol^{-1}
 C -176 kJ mol^{-1}
 D -214 kJ mol^{-1}

- 10 The following compound is an active component of 'tear gas'.

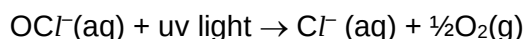


What are the numbers of σ and π bonds present in this molecule?

- | | σ | π |
|----------|----------|-------|
| A | 18 | 8 |
| B | 13 | 8 |
| C | 18 | 6 |
| D | 13 | 6 |
- 11 Which type of bonds / attractions is responsible for the intermolecular forces in liquid tetrachloromethane, CCl_4 ?
- A** covalent bonds
- B** hydrogen bonds
- C** induced dipole – induced dipole attractions
- D** permanent dipole – permanent dipole attractions
- 12 Swimming pool water can be kept free of harmful bacteria by adding aqueous sodium chlorate(I), NaOCl . This reacts with water to produce HOCl molecules which kill bacteria.



However in bright sunshine, the OCl^- ion is broken down by ultra-violet light.



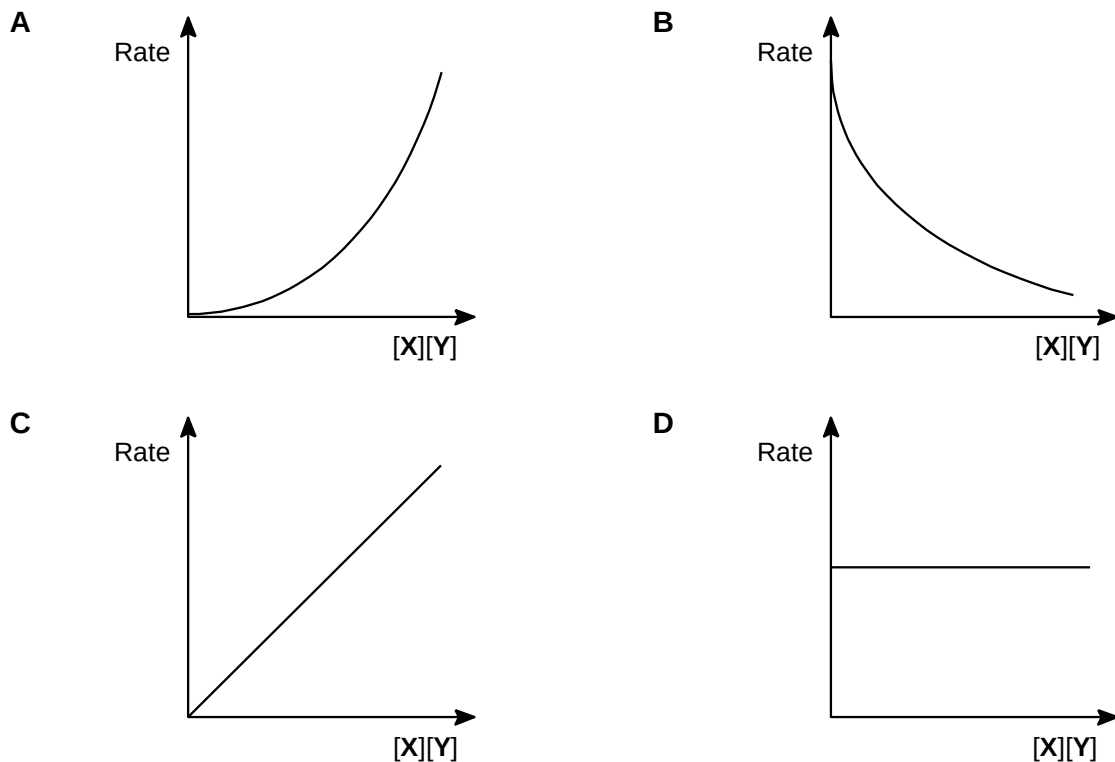
Which method would maintain the highest concentration of $\text{HOCl}(\text{aq})$?

- A** bubble air through the water
- B** add a solution of chloride ions
- C** add a solution of hydroxide ions
- D** acidify the pool water

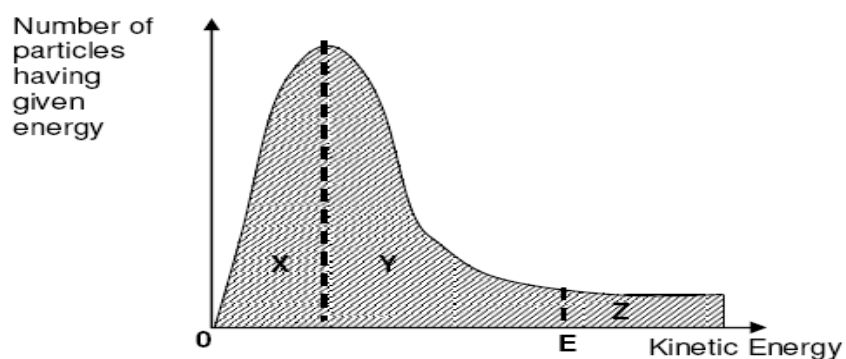
- 13 The reaction between X and Y is a second-order reaction.



The rate of this reaction is plotted against the product $[X][Y]$ of the concentrations of X and Y. Which graph would be obtained?



- 14 The Boltzmann distribution shows the number of molecules having a particular kinetic energy at constant temperature.



If the temperature is decreased by 20 °C, what happens to the size of the areas labelled X, Y and Z?

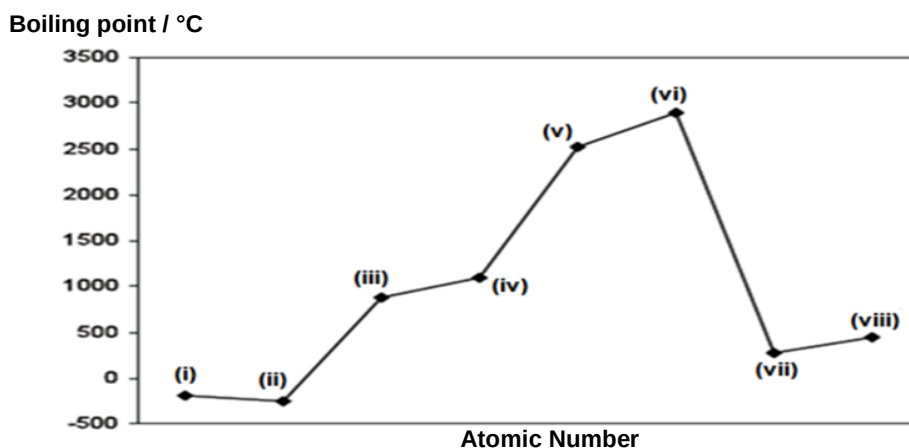
	X	Y	Z
A	Increase	Decrease	Increase
B	Increase	Decrease	Decrease
C	Decrease	Increase	Increase
D	Decrease	Increase	Decrease

- 15 Consecutive elements **P**, **Q** and **R** are in the third period of the Periodic Table. Element **Q** has the highest first ionisation energy and the lowest melting point.

What could be the identities of **P**, **Q** and **R**?

- A aluminium, silicon, phosphorus
- B magnesium, aluminium, silicon
- C silicon, phosphorus, sulfur
- D sodium, magnesium, aluminium

- 16 The following graph shows the boiling points of eight consecutive elements, **(i)** to **(viii)**, in Periods 2 and 3 of the Periodic Table.



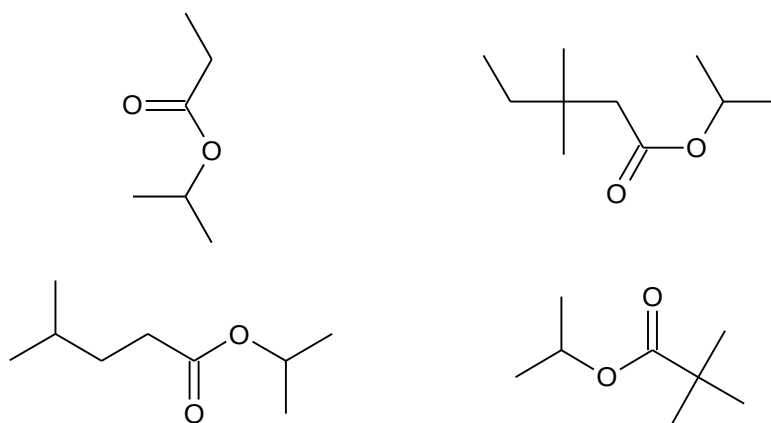
Which of the following statements about the oxides of the elements is **incorrect**?

- A **(vi)** forms a solid oxide.
- B **(v)** forms an oxide which can react with bases.
- C **(i)** forms an oxide with a melting point of 1286 °C.
- D **(viii)** can form an oxide with variable oxidation state.

- 17 10.0 cm³ of HCl solution of pH 1 and 10.0 cm³ of H₂SO₄ solution of pH 2 were mixed. Calculate the pH of the resulting mixture.

- A 1.0
- B 1.2
- C 1.3
- D 1.5

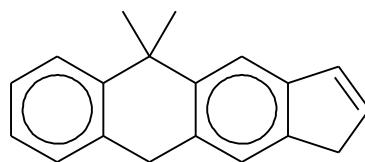
- 18** The following structures belong to some (non-consecutive) members of the aliphatic isopropyl ester homologous series.



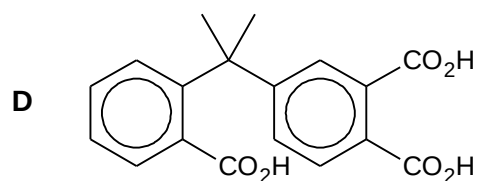
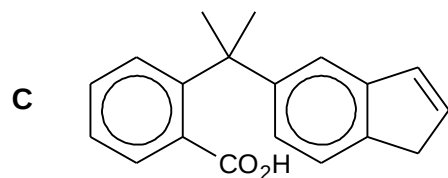
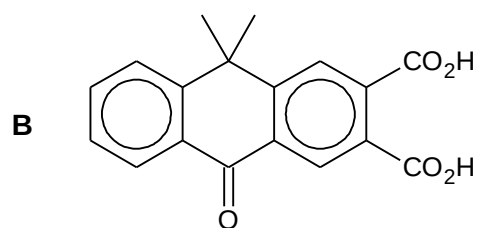
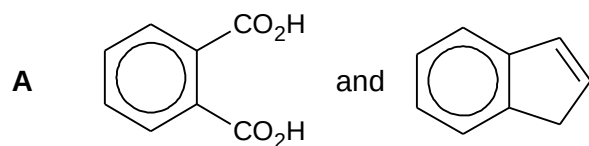
What is the general formula of this homologous series?

- A** $C_{n+3}H_{2n+8}O_2$
B $C_{n+4}H_{2n+8}O_2$
C $C_{n+3}H_{2n+6}O_2$
D $C_{n+4}H_{2n+6}O_2$
- 19** What is the total number of different chloroethanes, of formula $C_2H_{6-n}Cl_n$, where n can be any integer from 1 to 6?
- A** 6
B 8
C 9
D 10

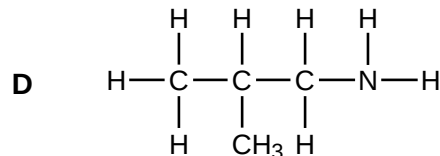
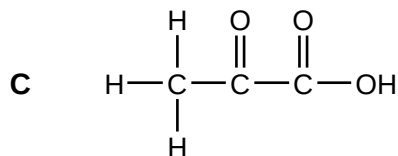
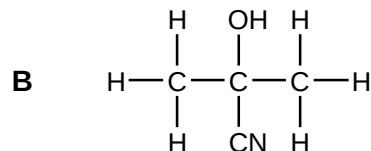
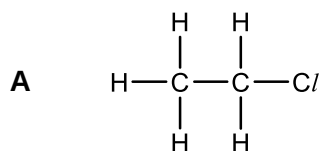
- 20 Compound **Z**, which has an aromatic ring structure, is subjected to oxidative degradation under suitable conditions.

**Z**

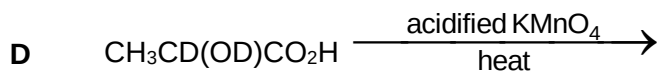
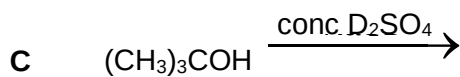
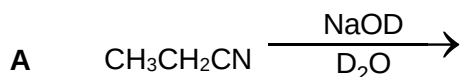
What are the most likely organic products from this reaction?



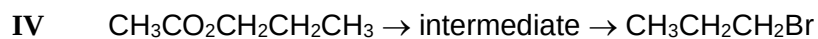
- 21 Which compound can be synthesised from prop-1-ene in less than 3 steps?



- 22 Which reaction yields a carbon compound incorporating deuterium, D? [D = ^2H]



- 23 Which pair of reactions could have the same intermediate?



A I and II

B I and III

C II and IV

D III and IV

- 24 Which compound is a product of the hydrolysis of $\text{CH}_3\text{CO}_2\text{C}_3\text{H}_7$ by boiling with aqueous sodium hydroxide?

A $\text{CH}_3\text{CO}_2\text{H}$

B $\text{C}_3\text{H}_7\text{OH}$

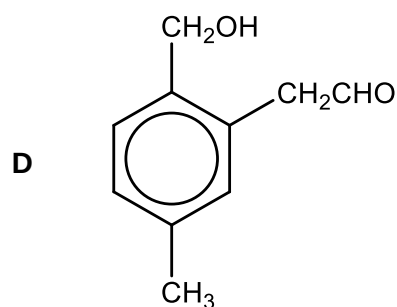
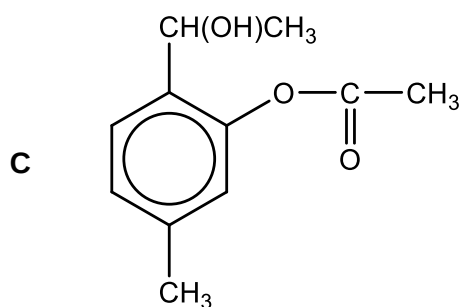
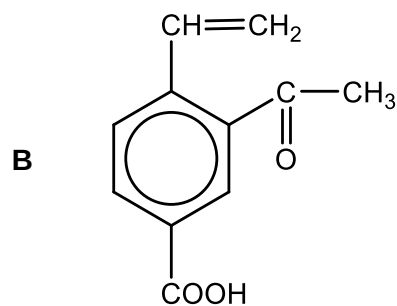
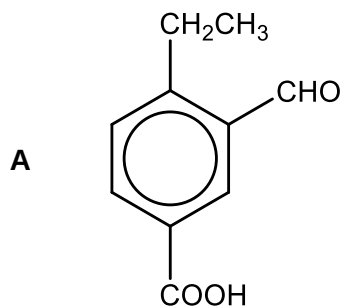
C $\text{C}_3\text{H}_7\text{O}^-\text{Na}^+$

D $\text{C}_3\text{H}_7\text{CO}_2^-\text{Na}^+$

25 Compound **Q** has the following properties:

- It produces white fumes when PCl_5 is added.
- It reacts with Na_2CO_3 to give rapid effervescence.
- It does not react with Cu^{2+} in alkaline solution but gives an orange precipitate with 2,4-DNPH.
- It results in the formation of green Cr^{3+} ions from an acidified solution of $\text{Cr}_2\text{O}_7^{2-}$ ions.

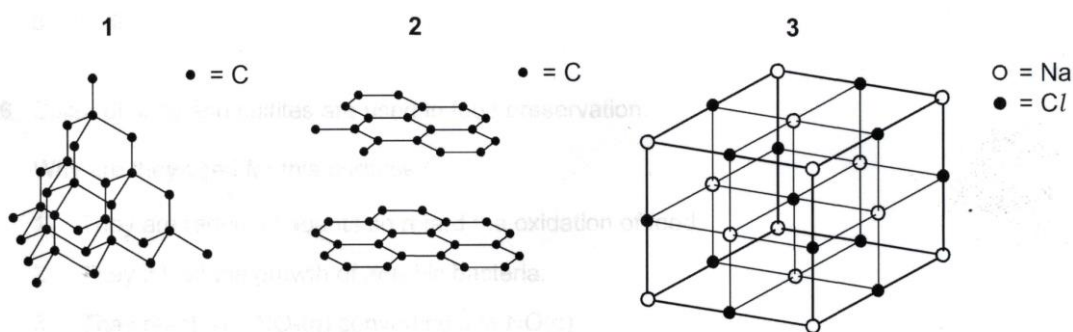
Which one of the following could be **Q**?



Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

26 Which diagrams represent part of a giant molecular structure?



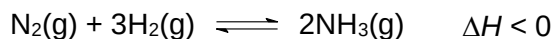
- Both H_2O and OF_2 molecules have similar bond angles.
- Boron nitride (BN) has a similar structure as graphite, and is thus a good conductor of electricity.
- AlF_3 and AlCl_3 are both simple molecular substances and form dimers at suitable temperatures.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 28** Ammonia is manufactured industrially by the Haber Process as shown.



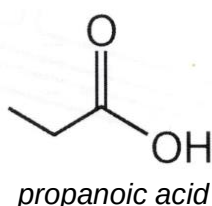
The operating conditions are:

400 to 450 °C; a pressure of 200 atm; an iron catalyst

Which of the following statements are **true** about the Haber process for the manufacture of ammonia?

- 1** At higher temperatures, the yield goes down but the rate of production of ammonia is faster.
- 2** At higher pressures, the yield goes down but the rate of production of ammonia is faster.
- 3** The presence of a catalyst shifts the equilibrium position to the right and increases the yield.

- 29** Propanoic acid occurs naturally as a result of the bacterial fermentation of milk, and is partly responsible for the flavour of Swiss cheese.



Which starting materials could be used to synthesise propanoic acid?

- 1** $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- 2** $\text{CH}_3\text{CH}_2\text{CN}$
- 3** $\text{CH}_3\text{CH}_2\text{CHO}$

- 30** Which of the following reactions are examples of substitution reactions?

- 1** $\text{CH}_3\text{CH}_2\text{Br} + \text{OH}^- \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{Br}^-$
- 2** $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{NH}_3 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 + \text{HCl}$
- 3** $\text{CH}_3\text{I} + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{CH}_3\text{OH} + \text{HI}$

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